

Tarun Reddy

Senior AI/ML & Generative AI Engineer

Email | Phone:

Professional Summary:

- Senior AI/ML & Generative AI Specialist with 4+ years** of hands-on experience building and deploying production-grade AI solutions across Healthcare and Financial Services domains. Strong background in designing scalable machine learning systems and delivering business-driven AI applications in regulated environments.
- Generative AI & LLMs:** Extensive experience developing RAG-based architectures, multi-agent workflows using LangGraph and CrewAI, and custom LLM integrations for use cases such as medical summarization and financial document automation.
- End-to-End ML Engineering:** Proven track record across the full ML lifecycle, including large-scale data processing with PySpark and Kafka, model development using PyTorch, TensorFlow, and Scikit-learn, and deploying robust solutions with MLOps best practices.
- Cloud & Scalable Systems:** Hands-on expertise in building and deploying AI platforms on AWS, Azure, and GCP, leveraging Docker, Kubernetes, and CI/CD pipelines to support high-availability, real-time inference services using FastAPI and Flask.
- Domain Expertise:**
 - ✓ **Healthcare:** Experience working with EHR/clinical data, implementing NLP-based classification models, and ensuring compliance with HIPAA and healthcare data standards.
 - ✓ **Financial Services:** Built fraud detection systems, trading sentiment models, and policy-driven predictive models in highly regulated financial environments.

Technical Skills:

Category	Skills
Programming Languages	Python, SQL, Shell Scripting
AI/Machine Learning	Generative AI, LLMs, RAG Pipelines, LangChain, LangGraph, CrewAI, TensorFlow, PyTorch, Scikit-learn, NLP, Transformer Models (BERT, GPT), Sentiment Analysis, Classification, Predictive Modeling
Backend& Frameworks	FastAPI, Django, Flask, REST APIs, Microservices Architecture
Cloud Platforms	AWS (EC2, S3, Lambda, SageMaker, IAM), Azure (Azure OpenAI, AKS), GCP (Vertex AI)
Data Engineering	Kafka, PySpark, Spark, Airflow, ETL/ELT Pipelines, Data Pipelines
Databases	PostgreSQL, Oracle, Hive, Redis
DevOps& Infrastructure	Docker, Kubernetes, Jenkins, Ansible, CI/CD Pipelines
Frontend	React, Angular, JavaScript, HTML, CSS

Education Details:

Bachelor's degree, Information Technology – JNTUH
Master's degree in science – Clark University

Professional Experience:

Client: Johnson & Johnson

Oct 2025 to Present

Role: Senior AI/ML Engineer (Python | GenAI)

Responsibilities:

- Developed and implemented Generative AI-based solutions leveraging LLMs to support medical document summarization, clinical knowledge extraction, and internal research use cases used by healthcare stakeholders.
- Built and deployed end-to-end machine learning models across multiple use cases including classification, NLP, and predictive analytics using TensorFlow, PyTorch, and Scikit-learn, primarily focused on patient risk analysis and clinical insights.
- Designed Redis-based caching and async processing, improving API throughput and reducing compute costs
- Developed LLM evaluation and prompt optimization frameworks, increasing response consistency and reducing hallucinations
- Hands-on experience in developing NLP-driven applications to process and extract meaningful insights from unstructured clinical notes, EHR data, and physician reports, enabling structured data generation for downstream systems.
- Integrated and customized LLM APIs (OpenAI and Azure OpenAI) into enterprise applications to build chatbot-based solutions supporting internal users, automating query handling, and improving accessibility to healthcare information.
- Engineered robust data pipelines using Python, Pandas, and PySpark, working with both structured and unstructured datasets, and ensured seamless data flow across cloud platforms such as AWS and Azure.
- Fine-tuned and optimized transformer-based models (BERT, GPT variants) for domain-specific healthcare applications including adverse event detection, clinical coding, and medical text classification tasks.
- Deployed containerized AI services on Azure AKS and GCP Cloud Run, improving scalability and system reliability across environments. Actively contributed to implementing MLOps practices, including CI/CD pipelines, model deployment, versioning, monitoring, and automated retraining workflows using MLflow, Docker, and Kubernetes in production environments.
- Architected RAG pipelines using vector databases (FAISS/Pinecone) and enterprise data sources, increasing contextual response accuracy for clinical use cases
- Focused on improving model performance and reliability through feature engineering, hyperparameter tuning, and applying explainability techniques such as SHAP and LIME to ensure transparency in predictions.
- Developed and deployed real-time inference systems and RESTful APIs using FastAPI and Flask, enabling seamless integration of ML models into enterprise healthcare applications.
- Ensured strict adherence to data privacy, security, and regulatory compliance standards (HIPAA/FDA) while working with sensitive healthcare data, implementing best practices for data governance and protection.

Environment: Python, Pandas, NumPy, PySpark, TensorFlow, PyTorch, Scikit-learn, NLP, Generative AI, LLMs, OpenAI API, Azure OpenAI, BERT, GPT, FastAPI, Flask, REST APIs, MLflow, Docker, Kubernetes, AWS, Microsoft Azure, CI/CD Pipelines, MLOps, SHAP, LIME, SQL, EHR Data Processing

Client: JPMorgan Chase & Co.

Dec 2024 to Oct 2025

Role: Python / AI-ML Engineer (GenAI, NLP, MLOps)

Responsibilities:

- Designed and implemented reusable GenAI wrappers and orchestration services in Python, allowing seamless switching between foundation models while maintaining consistent prompt engineering, response validation, and governance controls within a highly regulated financial environment.

- Developed and deployed AI-powered enterprise assistants to automate document drafting, internal knowledge retrieval, and business communication workflows, significantly reducing manual effort across operations, compliance, and research teams.
- Implemented CI/CD pipelines using Jenkins and Ansible, reducing deployment time and improving release reliability
- Built asynchronous APIs for large-scale financial workloads, improving system throughput and responsiveness
- Designed Power BI dashboards integrated with backend services, enabling data-driven decision-making for business teams
- Contributed to the enhancement of the OmniAI Machine Learning platform, building and scaling Python-based ML pipelines for real-time data processing and model inference, aligning with enterprise MLOps standards and governance frameworks.
- Implemented end-to-end MLOps workflows, including model training, validation, deployment, and monitoring using AWS and private cloud infrastructure, ensuring high availability, auditability, and compliance with internal risk policies.
- Built secure FastAPI/Django microservices for financial data processing with OAuth2-based authentication and RBAC
- Integrated Azure OpenAI services for real-time compliance monitoring and regulatory assistance workflows
- Built and maintained sentiment analysis models using advanced NLP techniques, leveraging alternative data sources such as financial news, earnings calls, and market signals to generate actionable insights for trading and research teams.
- Developed quantitative models such as SmartBuzz and BuzzMiner-style analytics, enabling large-scale ingestion and processing of unstructured text data to derive market sentiment and behavioral trends.
- Designed and implemented Hawk-Dove scoring models using NLP and time-series analysis to predict central bank policy direction based on historical meeting transcripts and macroeconomic indicators.
- Worked closely with cross-functional teams including data scientists, quants, and platform engineers to ensure seamless integration of AI/ML solutions into enterprise systems, adhering to strict security, compliance, and data governance standards.

Environment: Generative AI, LLMs, LangChain, LangGraph, CrewAI, Prompt Engineering, NLP, Machine Learning, TensorFlow, PyTorch, Scikit-learn, FastAPI, REST APIs, Microservices Architecture, AWS, MLOps, Model Deployment & Monitoring, , Data Pipelines, SQL, NoSQL, Docker, Kubernetes, CI/CD Pipelines, Unix/Linux, Time-Series Modeling, Enterprise AI Platforms, Risk & Compliance Frameworks

Client: Fortis Healthcare

Jan 2022 to Jul 2024

Role: Python Engineer

Responsibilities:

- Migrated legacy systems to Python-based microservices architecture, improving scalability and maintainability
- Optimized database queries (PostgreSQL, Hive), reducing processing time and improving system performance
- Deployed applications on AWS (EC2, S3, RDS), ensuring high availability and fault tolerance
- Built real-time monitoring solutions using WebSockets, improving system observability and incident response
- Helped with intent classification and entity extraction so that most of the common queries could be answered without agent involvement.
- Built a small pipeline to summarize call transcripts using transformer models, which made it easier for support teams to log call details.
- Integrated those summaries into the CRM system so agents didn't have to spend much time on manual notes.
- Used SHAP and LIME occasionally to understand model outputs and explain results to internal teams when needed.
- Deployed Python services using Docker and worked with Kubernetes for managing deployments in AWS.
- Used SageMaker for training and tracking models and helped with basic monitoring in production.

Environment: Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, Transformers (Hugging Face), , Anomaly Detection, Predictive Analytics, SHAP, LIME, Docker, Kubernetes, AWS, SageMaker, REST APIs, Data Pipelines, Microservices Architecture, Git